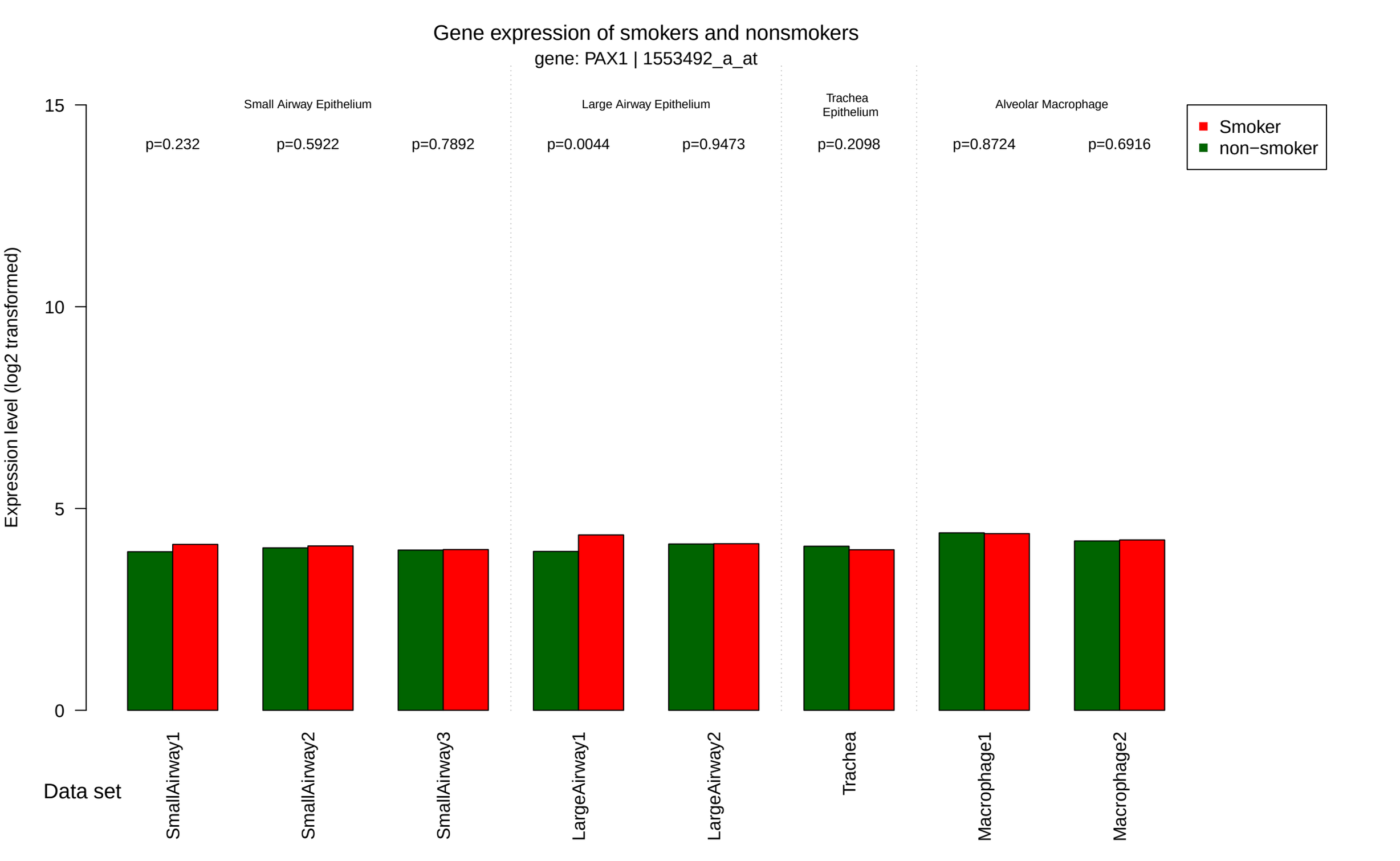
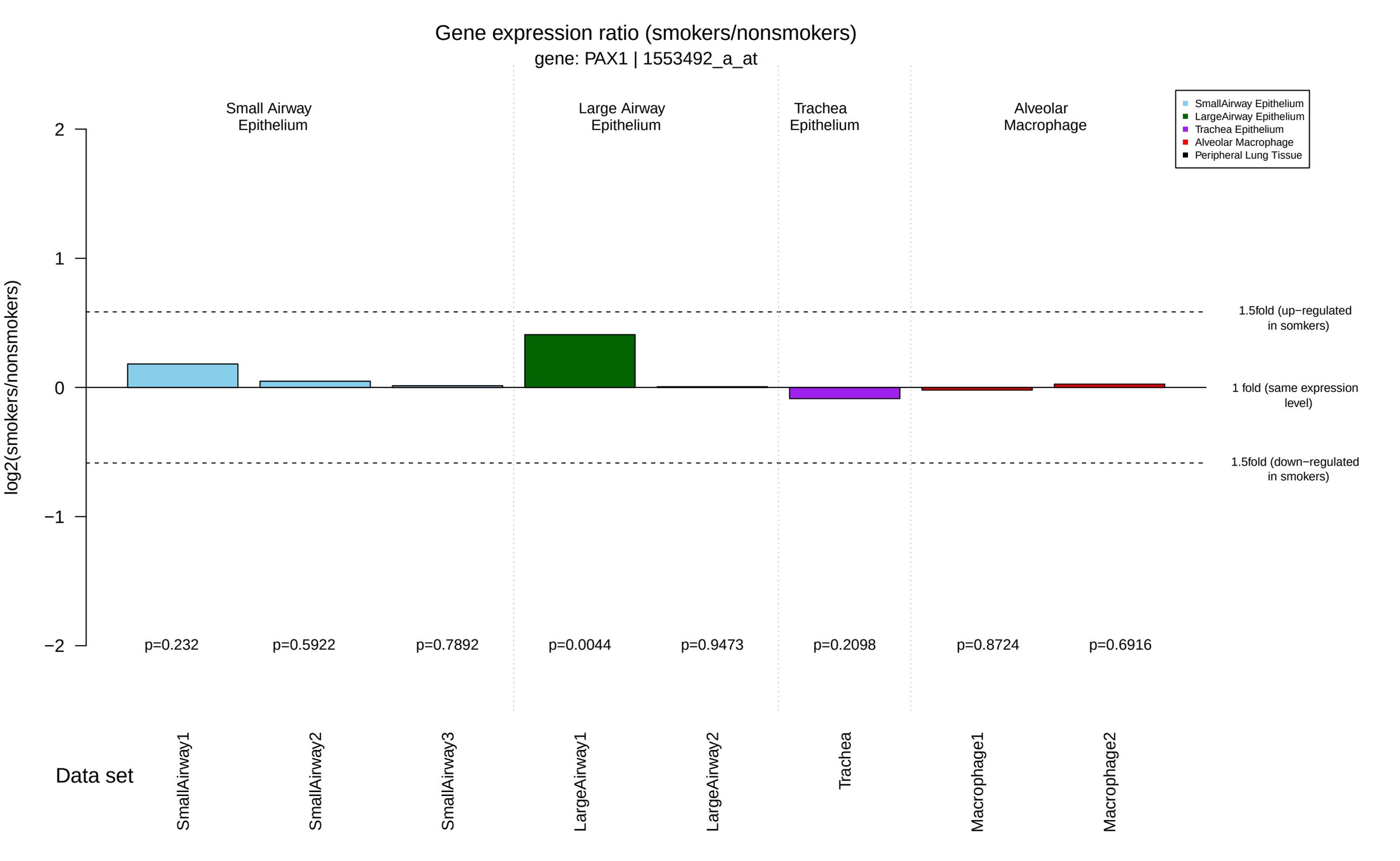
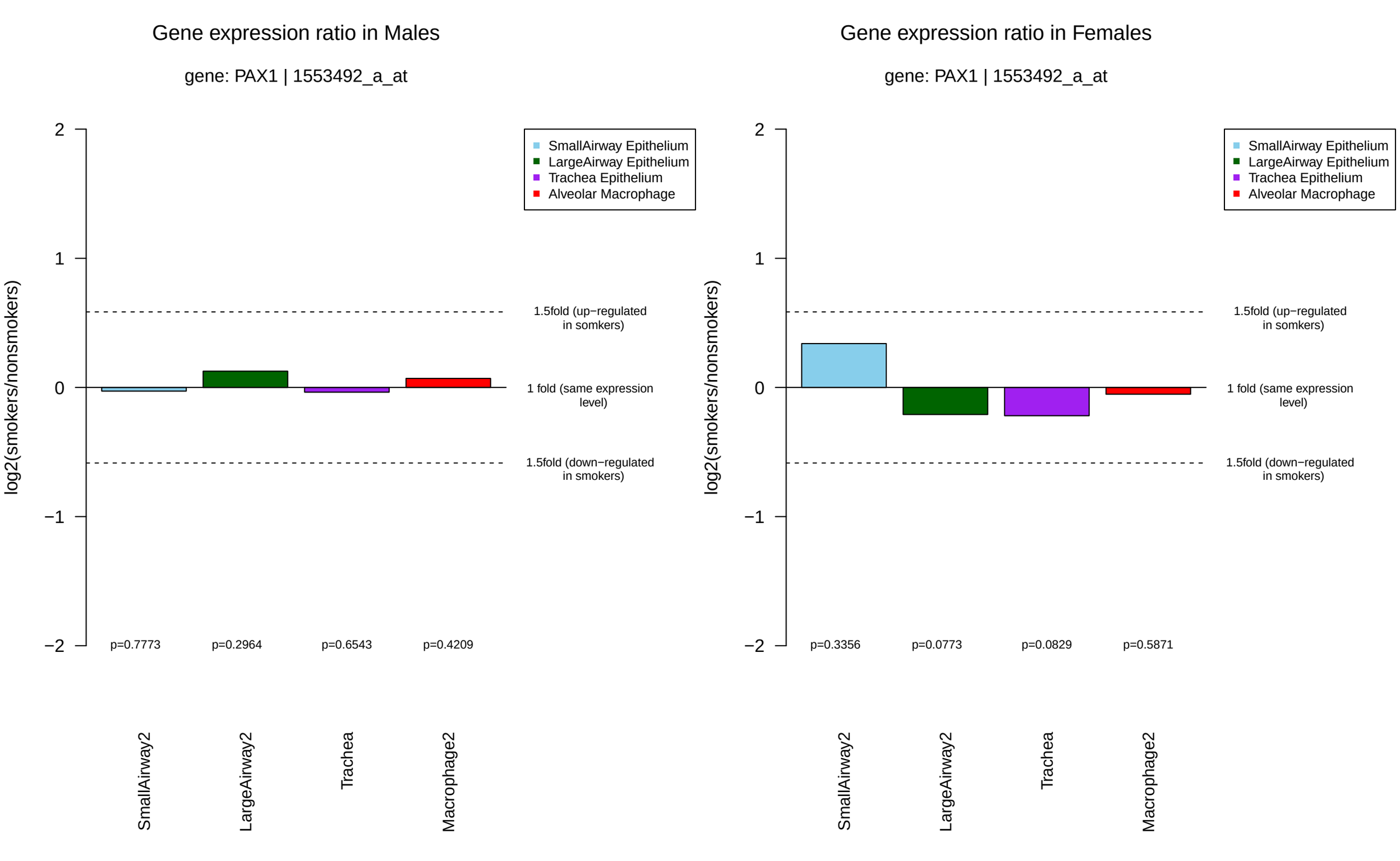
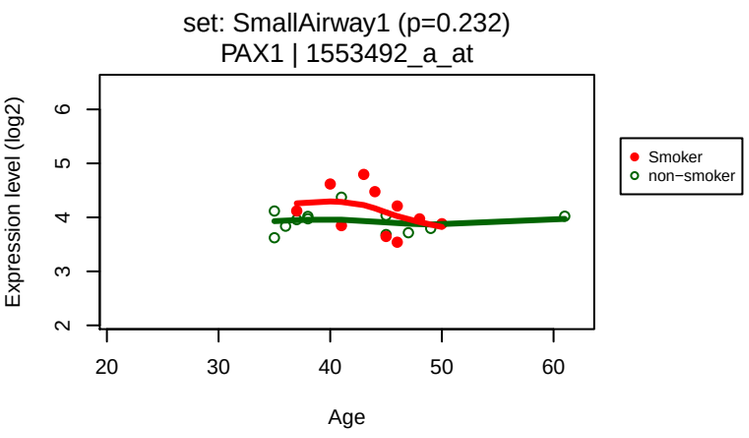
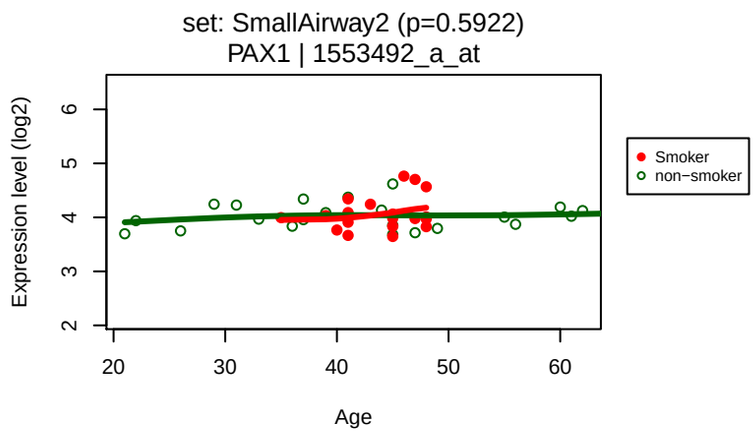
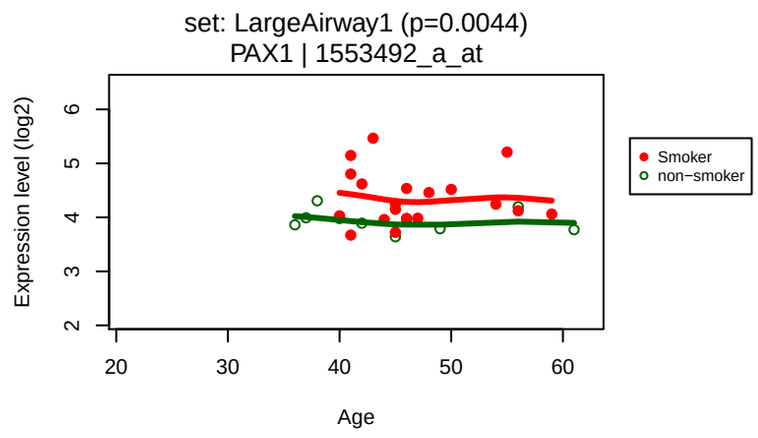

Gene expression of smokers and nonsmokers
gene: PAX1 | 1553492_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.232
p=0.5922
p=0.7892
p=0.0044
p=0.9473
p=0.2098
p=0.8724
p=0.6916
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

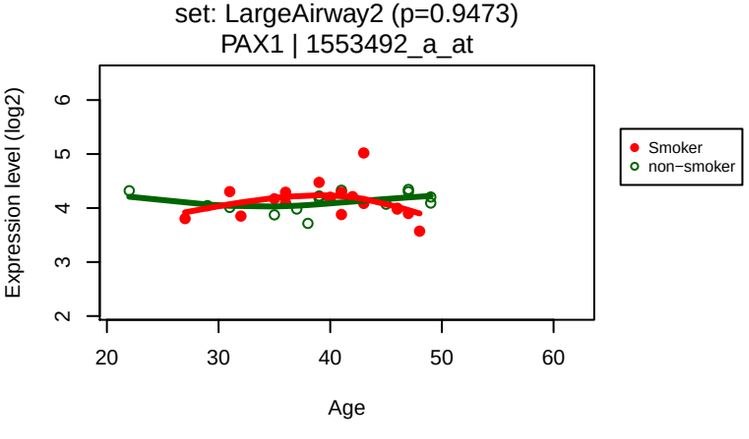

Gene expression ratio (smokers/nonsmokers)
gene: PAX1 | 1553492_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.232
p=0.5922
p=0.7892
p=0.0044
p=0.9473
p=0.2098
p=0.8724
p=0.6916
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

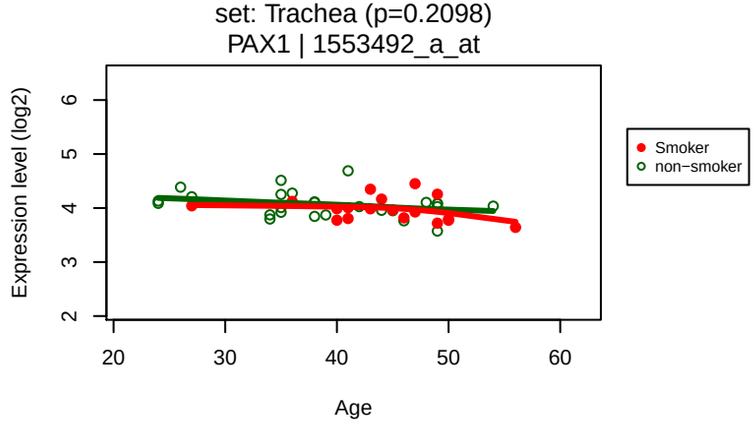

Gene expression ratio in Males
gene: PAX1 | 1553492_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7773
p=0.2964
p=0.6543
p=0.4209
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAX1 | 1553492_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3356
p=0.0773
p=0.0829
p=0.5871
SmallAirway2
LargeAirway2
Trachea
Macrophage2

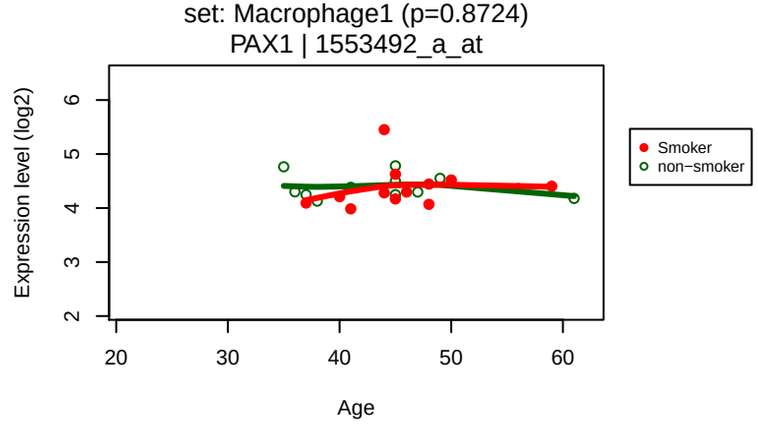

set: SmallAirway1 (p=0.232)
PAX1 | 1553492_a_at
Expression level (log2)
Age
Smoker
non−smoker

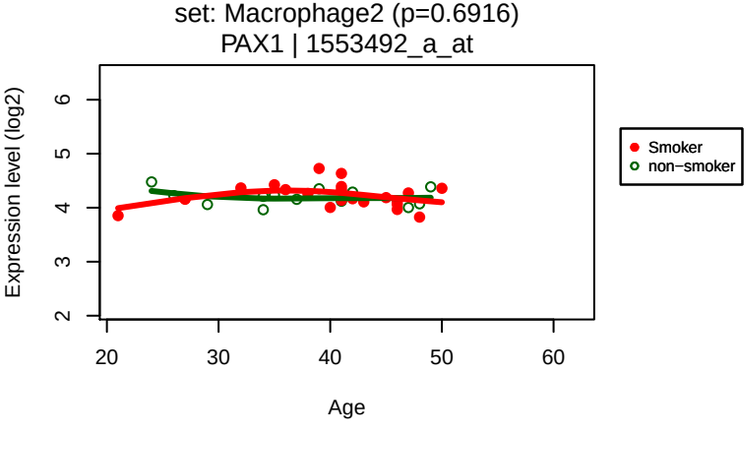

set: SmallAirway2 (p=0.5922)
PAX1 | 1553492_a_at
Expression level (log2)
Age
Smoker
non−smoker

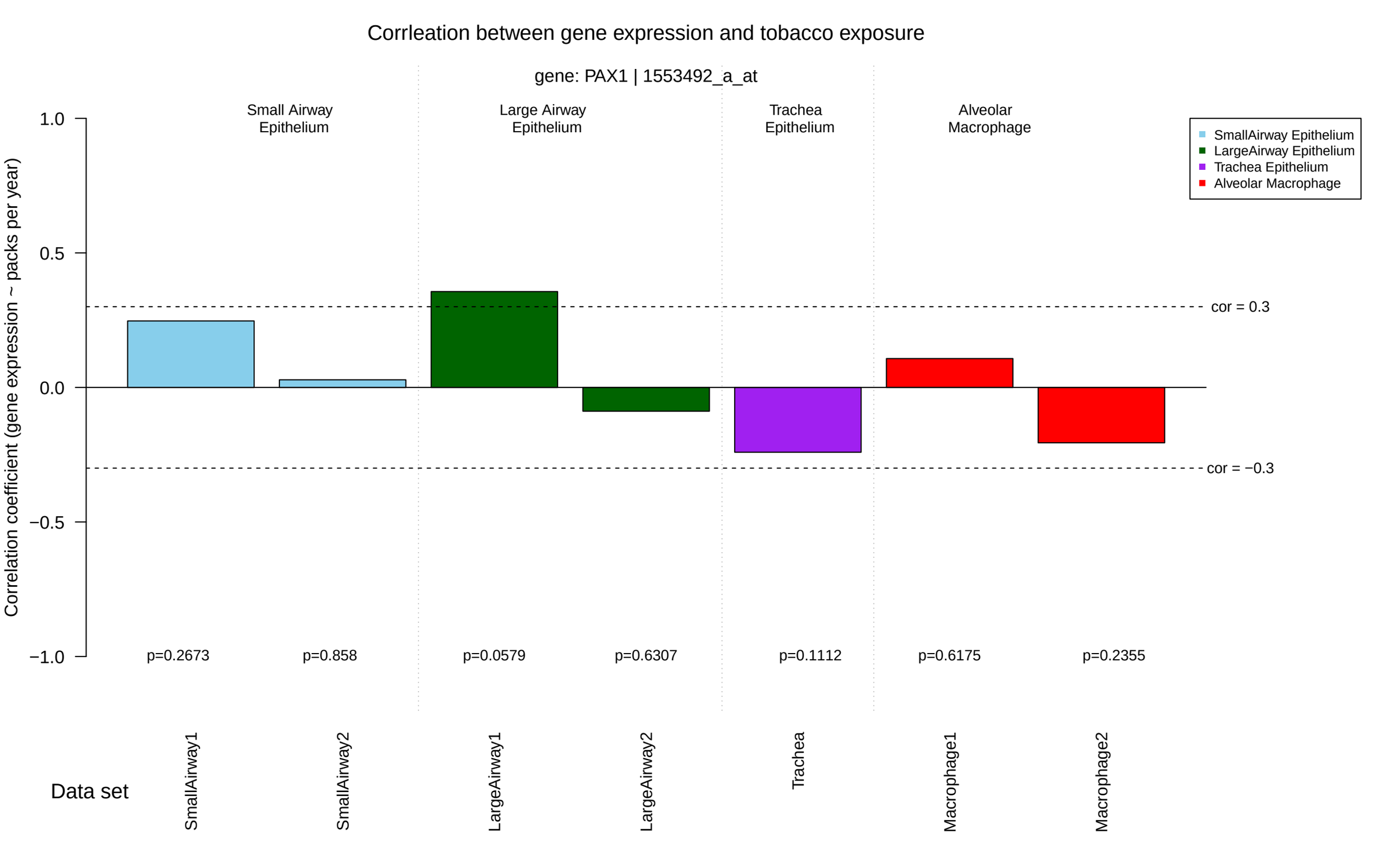

set: LargeAirway1 (p=0.0044)
PAX1 | 1553492_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9473)
PAX1 | 1553492_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2098)
PAX1 | 1553492_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8724)
PAX1 | 1553492_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6916)
PAX1 | 1553492_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAX1 | 1553492_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2673
p=0.858
p=0.0579
p=0.6307
p=0.1112
p=0.6175
p=0.2355
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2